

REVIEW

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# Persistent lead neurotoxicity as a consequence of global regulatory failure and trade-driven exposure inequities

Naveen Kumar Rajasekaran<sup>1</sup>, Chetan Ashok<sup>1</sup>, Srikanth Jeyabalan<sup>1,6\*</sup>, Gayathri Veeraraghavan<sup>2</sup>, Vetriselvan Subramaniam<sup>3</sup>, Devaraja Dravida Pandiyan<sup>1</sup>, Ling Shing Wong<sup>4</sup>, Mahendran Sekar<sup>5</sup> and Ravindran Jaganathan<sup>6</sup>

\*Correspondence:

Srikanth Jeyabalan

srikanth.j@sriramachandra.edu.in

<sup>1</sup>Department of Pharmacology, Sri Ramachandra Faculty of Pharmacy, Sri Ramachandra Institute of Higher Education and Research (DU), Chennai, Tamil Nadu 600 116, India

<sup>2</sup>Centre for Toxicology and Developmental Research (CEFTE), Sri Ramachandra Institute of Higher Education and Research (DU), Chennai, Tamil Nadu 600 116, India

<sup>3</sup>Department of Biomedical Sciences, Sir Jeffrey Cheah Sunway Medical School, Faculty of Medical and Life Sciences, Sunway University, Bandar Sunway, Petaling Jaya, Selangor Darul Ehsan 47500, Malaysia

<sup>4</sup>Faculty of Health and Life Sciences, INTI International University, Putra Nilai, Nilai, Negeri Sembilan 71800, Malaysia

<sup>5</sup>School of Pharmacy, Monash University Malaysia, Bandar Sunway, Selangor 47500 Subang Jaya, Malaysia

<sup>6</sup>Preclinical Department, Royal College of Medicine Perak, Universiti Kuala Lumpur (UniKL-RCMP), Ipoh, Perak 30450, Malaysia

## Abstract

**Background** The global phase-out of leaded gasoline, completed in 2021, is widely regarded as a major public health success. However, this achievement has fostered policy complacency, while legacy reservoirs and emerging exposure vectors continue to sustain a persistent neurotoxic burden. Low- and middle-income countries (LMICs) remain disproportionately affected due to enforcement constraints, informal recycling economies, and trade dynamics that externalise toxic risks.

**Objective** This critical narrative review integrates regulatory drivers of persistent lead exposure with molecular mechanisms of neurotoxicity to identify actionable leverage points for elimination.

**Methods** Peer-reviewed studies, surveillance reports, and case investigations published between 2015 and 2026 were synthesised. Evidence was organised across three domains: regulatory and enforcement gaps, contemporary environmental and consumer exposure pathways, and neurobiological mechanisms.

**Results** Lead exposure has shifted from historical fuel emissions to complex sources embedded in global trade and informal economies. Regulatory blind spots persist in high-income settings, while LMICs face enforcement challenges shaped by survival-based incentives. Exposure continues through contaminated soil, dust, water systems, food, and consumer products. Mechanistically, lead ( $Pb^{2+}$ ) acts as a calcium-mimetic neurotoxicant that crosses the blood-brain barrier, disrupts NMDA-dependent synaptic plasticity, and promotes mitochondrial dysfunction and apoptosis, contributing to long-term cognitive and economic harm.

**Conclusion** Persistent lead neurotoxicity reflects governance failure rather than scientific uncertainty. Eliminating exposure requires closing export loopholes, strengthening enforcement, formalising informal sectors through realistic incentives, and shifting from secondary screening to primary prevention.

**Keywords** Lead exposure, Neurotoxicity, Blood lead level, Regulatory gaps, Avgas, Informal recycling, Lead chromate, NMDA receptor, Mitochondria, SDG 3 - Public health



## 1 Introduction: the unfinished agenda

### 1.1 The global paradox

A ‘Solved’ Problem? The phasing out of leaded gasoline, completed globally in 2021, stands as a major public health achievement. Yet, it has fostered policy complacency, with the prevailing assumption that the crisis of environmental lead exposure has been resolved [1]. In reality, legacy lead from decades of vehicular emissions persists in urban soils, continually remobilising into the atmosphere and perpetuating toxicity in modern populations [1]. This ongoing exposure imposes a staggering economic toll: contemporary childhood lead exposure is estimated to cost the global economy \$3.4 trillion USD annually, a figure equivalent to approximately 2% of global GDP [2]. Crucially, this economic shock is unevenly distributed, with low- and middle-income countries (LMICs) bearing the heaviest burden of productivity losses and healthcare costs [2]. From a clinical perspective, lead remains a ‘non-threshold toxicant’ with no established safe level of exposure [1, 3]. Beyond the well-documented neurodevelopmental deficits in children, chronic exposure is now implicated in the pathogenesis of neurodegenerative conditions in ageing populations, including cognitive decline and an increased risk of Alzheimer’s disease [3, 4].

### 1.2 The shifting landscape: From Tailpipes to ‘Trade and Trash’

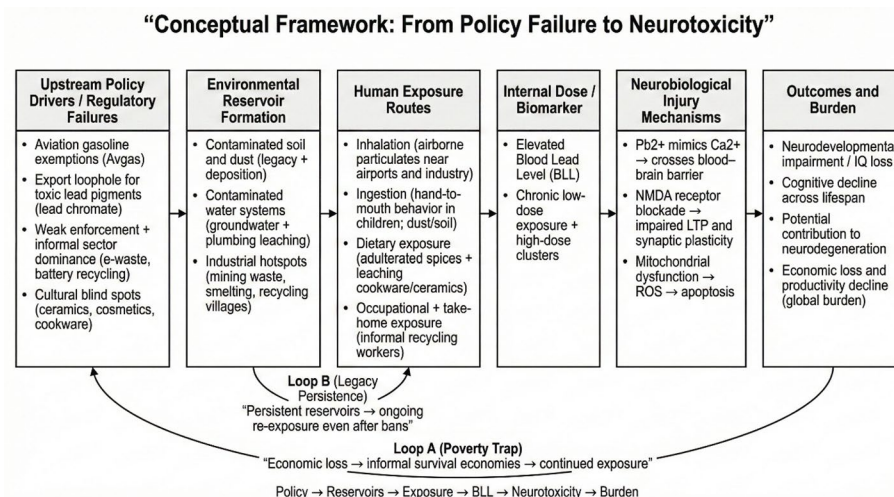
The era of leaded petrol has closed, but the vectors of neurotoxicity have shifted from tailpipes to a fragmented, unregulated web of global trade and waste management. The explosive growth of the informal e-waste recycling sector has become a primary driver of exposure, releasing hazardous lead compounds into the environment, particularly across Asia and Africa [5]. Simultaneously, a global trade-driven risk externalisation has emerged wherein high-income nations, despite strictly regulating or banning lead chromate domestically, continue to export these toxic pigments to countries with weaker regulatory frameworks [6].

Furthermore, under-recognised lead contamination has been detected in essential consumer goods; a rapid market screening across 25 LMICs found lead levels exceeding safety reference values in 18% of samples, including metal foodware, ceramics, and spices [7]. This risk is exemplified by the adulteration of turmeric with lead chromate pigments, a practice driven by market incentives that threatens public health across South Asia [8].

To integrate these upstream regulatory contradictions with downstream biological injury and societal impacts, we propose a conceptual framework linking policy failure to environmental reservoir formation, human exposure routes, internal dose (blood lead level), lead-induced neurotoxicity mechanisms, and long-term outcomes (Fig. 1).

Therefore, this review synthesises how modern lead exposure persists through interconnected regulatory contradictions, informal economic drivers, and legacy environmental reservoirs. We integrate these upstream governance failures with downstream biological mechanisms of neurotoxicity and population-level economic consequences. Finally, we propose actionable prevention priorities to shift global strategies from secondary screening to primary prevention and elimination.

Regulatory gaps and enforcement failures drive environmental reservoir formation (soil/dust/water), sustaining exposure via inhalation, oral ingestion (dietary intake and incidental hand-to-mouth ingestion), dermal/mucosal contact, and occupational



**Fig. 1** Conceptual framework linking regulatory failure to persistent lead exposure and neurotoxicity

inhalation routes, resulting in elevated blood lead levels (BLL). Pb<sup>2+</sup> neurotoxicity occurs through calcium mimicry, NMDA/LTP disruption, and mitochondrial dysfunction, leading to neurodevelopmental impairment and economic burden, reinforced by feedback loops of poverty and legacy persistence.

## 2 Methods

### 2.1 Study design

This article is a critical narrative review with a policy to mechanism perspective, designed to synthesise contemporary evidence on why lead exposure persists despite decades of regulatory progress and scientific clarity regarding lead neurotoxicity.

### 2.2 Literature search strategy

A targeted literature search was conducted to identify peer-reviewed evidence describing (i) modern lead exposure vectors, (ii) regulatory gaps and enforcement barriers, (iii) environmental reservoirs and consumer product sources, and (iv) mechanistic neurotoxic outcomes. The search strategy focused primarily on literature published between 2015 and 2026, with earlier landmark references included where necessary to support foundational concepts. PubMed, Scopus, Web of Science, and Google Scholar were searched for relevant publications.

Search terms included combinations of: “lead exposure,” “blood lead level,” “neurotoxicity,” “LMIC,” “informal recycling,” “e-waste,” “used lead acid batteries,” “lead chromate,” “turmeric adulteration,” “consumer products,” “kohl,” “ceramics,” “cookware,” “aviation gasoline,” “airport lead emissions,” “soil lead,” “legacy mining,” “Kabwe,” “Zamfara,” “policy,” “regulatory enforcement,” “trade export loophole,” “primary prevention,” and “public health intervention.”

### 2.3 Eligibility and study selection

A total of 156 records were identified, of which 135 remained after duplicate removal and were screened by title/abstract; 59 were included for qualitative synthesis using predefined inclusion criteria addressing contemporary exposure vectors, measurable contamination or biomonitoring outcomes, regulatory evidence, and mechanistic

neurotoxicity. Evidence was included if it met one or more of the following criteria: (i) quantitative documentation of lead exposure or blood lead levels in communities, workers, or children; (ii) environmental measurements of lead contamination in soil, dust, water, food, or consumer products; (iii) mechanistic evidence describing neurobiological effects of  $\text{Pb}^{2+}$  across cellular, synaptic, mitochondrial, and neurodevelopmental pathways; and/or (iv) case studies, interventions, or policy analyses describing regulatory success, failure, or enforcement barriers. To minimise selection bias, priority was given to primary data sources and high-authority reports, with case studies selected to represent recurring exposure contexts across geographic and economic settings. Studies were excluded if they did not address lead exposure, lacked measurable exposure outcomes, or were not aligned with the mechanistic or policy objectives of this review.

## 2.4 Data synthesis

The included evidence was synthesised thematically and organised into a three-layer framework:

- Global regulatory paradoxes (policy contradictions, export loopholes, enforcement failures, and informal-sector dynamics).
- Environmental reservoirs and exposure routes (soil/dust, water systems, food chain and consumer vectors).
- Neurobiological mechanisms and systemic outcomes (BBB transport, NMDA/LTP disruption, mitochondrial dysfunction, apoptosis, and broader health/economic burden).

Findings were presented through structured case examples and summarised using two integrative tables: (i) a global matrix of regulatory paradoxes and exposure vectors (Table 1), and (ii) mechanistic pathways linking  $\text{Pb}^{2+}$  exposure to neurotoxicity (Table 2).

## 3 Conceptual framework: from policy failure to neurotoxicity

To link governance failures to measurable neurological harm, this review proposes an integrated causal framework in which regulatory decisions and enforcement capacity shape exposure environments, thereby determining internal dose, mechanistic injury, and population-level outcomes. The framework follows a sequential pathway:

(1) regulatory failure → (2) environmental reservoir formation → (3) human exposure routes → (4) internal dose (blood lead level; BLL) → (5) molecular neurotoxicity → (6) cognitive and systemic impairment → (7) economic burden and intergenerational poverty.

First, regulatory contradictions in high-income countries (e.g., continued exemptions for leaded aviation gasoline and permissive export structures for hazardous pigments) and enforcement failures in LMIC settings (e.g., informal battery recycling and consumer product adulteration incentives) function as upstream policy drivers that sustain environmental contamination. These governance failures continuously replenish lead reservoirs in soil and dust, contribute to contamination of water systems through legacy mining waste and infrastructure leaching, and introduce lead into the food chain through both environmental uptake and intentional adulteration.

Second, sustained exposure occurs through multiple routes, including inhalation of resuspended dust particles, oral ingestion (comprising dietary intake from contaminated food and cookware leaching, and incidental non-dietary ingestion via hand-to-mouth behaviour in children), and dermal or mucosal exposure via cosmetics and traditional preparations. Together, these pathways generate a chronic, multi-source exposure profile that is often under-recognised by conventional surveillance systems.

Third, at the mechanistic level,  $Pb^{2+}$  acts as a calcium-mimetic neurotoxicant capable of bypassing protective barriers and disrupting synaptic and mitochondrial integrity.  $Pb^{2+}$ -mediated impairment of NMDA receptor signalling compromises long-term potentiation and neuroplasticity, while mitochondrial injury promotes oxidative stress, bioenergetic dysfunction, and apoptosis. These cellular injuries contribute to clinically meaningful cognitive decline and neurological dysfunction, with emerging evidence supporting long-term neurodegenerative risk.

Finally, the resulting population-level impairment reduces educational attainment, productivity, and lifetime earnings, thereby reinforcing socioeconomic inequality and constraining national development. Overall, this conceptual framework supports the central conclusion of this review: persistent lead neurotoxicity in the modern era reflects a failure of governance and implementation rather than a failure of scientific understanding.

## **4 Global regulatory paradoxes: regulatory contradictions and implementation gaps**

### **4.1 Regulatory contradictions in high-income nations**

Despite the ostensible eradication of lead from automotive gasoline and residential paints in the Global North, a critical examination of current policy reveals a persistent regulatory paradox wherein economic considerations sustain significant toxic “blind spots.” In the United States, this regulatory contradiction is most evident in the “Aviation Gap,” where piston-engine aircraft continue to utilise leaded aviation gasoline (avgas), creating a persistent reservoir of neurotoxic exposure. The U.S. Environmental Protection Agency (EPA) estimates that approximately 5.2 million people reside within 500 m of an airport where avgas is used, placing large surrounding populations at risk of chronic exposure [9]. This exposure results in measurable physiological harm; a comprehensive analysis of children residing near Reid-Hillview Airport in California demonstrated a clear, monotonically decreasing dose-response relationship between residential proximity to the airport and elevated blood lead levels (BLLs), with traffic volume serving as a significant predictor of toxicity [10]. Furthermore, the transition to unleaded alternatives is frequently stalled by economic and logistical justifications. For instance, in Alaska, the reliance on piston-engine aircraft for essential supply chains to off-road communities has been used to justify legislative amendments extending the transition goal to as late as 2032, prioritising supply chain stability over immediate abatement of lead emissions [40].

A parallel form of regulatory negligence exists in the international trade policies of high-income nations, constituting an “Export Loophole.” While the European Union and other wealthy nations have strictly prohibited or restricted the domestic use of lead chromate pigments due to their known toxicity, they continue to facilitate the export of these hazardous materials to countries with weaker regulatory capacities. Todd et al.

**Table 1** Global matrix of regulatory paradoxes, exposure vectors, and public health impacts (2015–2026)

| Region / country           | Main population            | Evidence type                            | Primary exposure vector             | Regulatory failure (Mechanism)                                                                                                       | Quantitative impact / Key Finding                                                                                                                        | Detected levels / toxic dose                                                                                                      | References |
|----------------------------|----------------------------|------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------|
| North America (USA)        | Children (Near Airports)   | Biomonitoring & Spatial Analysis         | Aviation Fuel (Avgas) & Legacy Dust | Regulatory Asymmetry: Lead is banned in cars (1996) but legal in 170,000 piston-engine aircraft; legacy urban soil remains untreated | BLLs in children living downwind of airports are dose dependently higher; 5.2 million people are exposed to aviation dust                                | Soil: Exceeds EPA Hazard Threshold (> 400 mg/kg)<br>Blood: Paediatric BLLs surpass CDC Reference Value (> 3.5 µg/dL)              | [1, 9, 10] |
| Europe (EU / UK)           | General Population (LMICs) | Policy Analysis & Trade Data             | Toxic Exports (Lead Chromate)       | The Export Loophole: REACH regulations ban domestic use of Lead Chromate but permit export to LMICs                                  | > 242 tons of lead pigment are exported annually (e.g., via Spain) to countries with weak surveillance                                                   | Pigment: Industrial Lead Chromate concentrate exceeds 10,000 mg/kg (10%)                                                          | [6, 11]    |
| Middle East (Saudi Arabia) | Women & Children           | Environmental Sampling (Product Testing) | Cosmetics (Kajal & Modern)          | Import Control Failure: Cultural use of Kohl combined with unregulated "low-price" commercial cosmetics                              | A significant lead was detected in traditional Kohl and modern lipsticks/foundations; contamination persists despite high GDP                            | Kohl: Traditional samples contain > 50% lead (> 500,000 mg/kg).<br>Cosmetics: Commercial samples exceed the 20-mg/kg safety limit | [12–14]    |
| Latin America (Mexico)     | Rural Communities          | Biomonitoring & Leachate Testing         | Traditional Glazed Ceramics         | Cultural Blind Spot: Traditional low-temperature glazed pottery leaches lead into acidic foods (salsa/stew)                          | 25% of children in rural areas have BLLs > 5 µg/dL, significantly associated with use of traditional glazed pottery; remediation is culturally resistant | Leachate: Foodware releases > 2–5 mg/kg into food<br>Blood: Chronic BLLs range from 5–20 µg/dL                                    | [15, 16]   |
| Asia (India)               | Informal Workers           | Policy Analysis & Soil Sampling          | Informal Battery Recycling (ULAB)   | The Implementation Gap: Laws exist (Battery Rules 2022), but a "parallel economy" uses counterfeit certificates                      | 90% of Used Lead Acid Batteries (ULABs) processed informally; soil levels near recycling sites exceed 400 mg/kg                                          | Soil: Extreme contamination (3,000–100,000 mg/kg) near informal smelters                                                          | [17–19]    |
| Asia (Vietnam)             | Children (Craft Villages)  | Biomonitoring (Cohort Study)             | "Craft Villages" & E-Waste          | Economic Disincentive for Compliance: Recycling occurs in residential homes; enforcement is lax to protect rural employment          | 100% of children in Dong Mai village exceeded 10 µg/dL; E-waste processing releases heavy metal cocktails                                                | Blood: 28% of children had BLL ≥ 45 µg/dL (Level requiring immediate chelation therapy)                                           | [5, 20–22] |
| Africa (Nigeria)           | Children                   | Biomonitoring & Mortality Data           | Gold Ore Processing & Paint         | Acute Extraction: Artisanal gold mining releases lead dust; new paints still contain lead                                            | Approximately 400 children died in Zamfara (2010–2013); new solvent-based paints exceed the 90-mg/kg limit                                               | Ore Dust: Processed dust exceeds 10,000 mg/kg.<br>Paint: Solvent-based brands exceed 90 mg/kg limit                               | [23, 24]   |



Table 1 (continued)

| Region / country                       | Main population    | Evidence type          | Primary exposure vector      | Regulatory failure (Mechanism)                                                                                                      | Quantitative impact / Key Finding                                                                     | Detected levels / toxic dose                                                                                                                                                              | References  |
|----------------------------------------|--------------------|------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Africa (Zambia)                        | General Population | Environmental Sampling | Legacy Mine Waste            | Legacy mine contamination: Mines closed decades ago (Kabwe) lack remediation; groundwater and residential areas remain contaminated | 100% of tested boreholes exceeded WHO safety limits; remediation attempts have faced re-contamination | Soil: Residential soil exceeds 2,500 mg/kg.<br>Dust: Indoor dust exceeds 1,000 mg/kg.<br>Blood Lead Levels: 1.65–162 µg/dL; 23% of children ≥ 45 µg/dL; highest township means 45.7 µg/dL | [25, 26]    |
| Global (Various)                       | Consumers          | Market Surveillance    | Consumer Goods (Spices/Toys) | The Supply Chain Gap: Lead found in toys, spices, and cosmetics across 25 LMICs.                                                    | 18% of consumer goods in LMICs contained > 100 mg/kg lead; spices are a major global vector           | Spices: Adulterated Turmeric contains 100–1152 mg/kg (Lead Chromate)                                                                                                                      | [7, 8]      |
| Asia (Bangladesh & broader South Asia) | General Population | Intervention Study     | Turmeric Adulteration        | The Success Model: Strict enforcement (fines) + Public Education campaigns                                                          | Success: Reduced lead-tainted turmeric from 47% to 0%; workers' BLLs dropped by 30%                   | Pre-Intervention: > 1,000 mg/kg.<br>Post-Intervention: < 2 mg/kg (Undetectable).                                                                                                          | [8, 27, 28] |

Values reported across studies are not directly comparable due to substantial heterogeneity in sampling methodologies, population demographics, exposure metrics, and analytical endpoints; cross-study comparisons should be interpreted with caution

**Table 2** Molecular mechanisms of lead-induced neurotoxicity

| Target system       | Molecular target                      | Mechanism of action                                                                                                  | Physiological consequence                                                                       | References   |
|---------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------|
| Blood-Brain Barrier | Voltage-Gated Calcium Channels (VGCC) | Ionic Mimicry: Pb <sup>2+</sup> (1.19 Å) mimics Ca <sup>2+</sup> (1.00 Å) and disrupts calcium transport mechanisms. | Lead accumulation in astrocytes and neurons; disruption of intracellular calcium homeostasis.   | [29–31]      |
| Synapse (LTP)       | NMDA Receptor (GluN2A)                | Receptor Blockade: Lead binds to the Zinc (Zn <sup>2+</sup> ) regulatory site, physically blocking the channel pore. | Inhibition of calcium influx; suppression of Long-Term Potentiation (LTP) and memory formation. | [32–34]      |
| Signaling Cascade   | CaMKII / CREB / BDNF                  | Signal Failure: Blocked NMDA receptors fail to activate CaMKII and CREB; suppression of BDNF expression.             | Retraction of dendritic spines; failure of gene transcription required for synaptic plasticity. | [33, 35, 36] |
| Mitochondria        | Mitochondrial Calcium Uniporter (MCU) | Energy Crisis: Lead-induced mitochondrial dysfunction associated with ETC impairment and ROS generation.             | ATP depletion; generation of Reactive Oxygen Species (ROS); metabolic collapse.                 | [37–39]      |
| Cell Survival       | DRP1 / mPTP / cytochrome c            | Apoptosis: ER stress upregulates DRP1 → Mitochondrial fragmentation → cytochrome c release.                          | Activation of the Caspase cascade; neuronal cell death; link to Alzheimer’s pathology.          | [3, 4, 37]   |

Values reported across studies are not directly comparable due to substantial heterogeneity in sampling methodologies, population demographics, exposure metrics, and analytical endpoints; cross-study comparisons should be interpreted with caution

(2025) document that high-income jurisdictions, including the United States and Canada, continue to export significant quantities of lead chromate, often via transit hubs like Spain, to low- and middle-income countries, effectively externalising the toxic burden they refuse to tolerate domestically [6]. This creates a circular contamination pathway, whereby hazardous materials exported from high-income nations re-enter global food supply chains, ultimately compromising food safety in the exporting countries themselves. Sahoo and Samal (2024) highlight that adulterated spices, often contaminated with bright pigments like lead chromate to mask inferior quality, permeate global supply chains, thereby reintroducing these hazards into the very nations that exported the raw toxic materials [11].

4.2 Regulatory implementation gaps in low- and middle-income countries

In developing nations, the primary challenge is not the absence of regulation, but a profound “Implementation Gap” where laws exist on paper but are systematically ignored by a dominant informal sector driven by poverty. In India, this gap is characterised by the circulation of “counterfeit e-waste certificates,” which allow informal operators to bypass Extended Producer Responsibility (EPR) mandates. Heinrich et al. (2025) reveal that informal collectors in cities like Chandigarh utilise fake affidavits to bid for tenders, undermining the formal recycling ecosystem and perpetuating unregulated processing [17]. This regulatory failure sustains a “parallel economy” of backyard recycling, as described by Sharma and Pandey (2020) in the Mayapuri scrap market. Here, the dismantling of End-of-Life Vehicles (ELVs) occurs in unorganised, crude settings where economic subsistence drives workers to process hazardous materials without protective equipment or environmental controls, prioritising material recovery over safety standards [18].



Vietnam illustrates a distinct manifestation of this gap through the “Craft Village” model, where rural industrialisation policies have embedded toxic processing within residential communities. Dang and Tran (2020) note that while these villages generate employment, they operate with outdated technology and limited infrastructure, leading to severe environmental degradation [20]. The consequences of this model are starkly evident in Dong Mai, a battery recycling village where entire families process batteries in their homes. Daniell et al. (2015) found that despite government efforts to relocate these activities, 28% of children in the village had blood lead levels  $\geq 45 \mu\text{g/dL}$ , demonstrating how the convenience and economic necessity of home-based work override regulatory attempts to separate industrial hazards from living spaces [21]. While this study dates to 2015, subsequent investigations confirm that informal lead recycling continues to operate in comparable settings across Vietnam and the broader Southeast Asian region, sustaining similar contamination profiles in the absence of effective regulatory enforcement.

#### 4.3 The resource curse: extraction and legacy

The burden of lead toxicity in the Global South and resource-rich regions is further compounded by the “Resource Curse,” manifesting as both acute disasters from artisanal extraction and chronic exposure from industrial legacies. In Nigeria, the catastrophic consequences of unregulated extraction were realised in the Zamfara gold mining crisis, where artisanal processing of lead-rich ore released massive quantities of dust, causing an acute poisoning event that killed hundreds of children [23]. In contrast, Zambia faces a “chronic” legacy crisis in Kabwe, where the historical footprint of lead mining continues to contaminate groundwater decades after mine closure. Siame et al. (2025) report that 100% of borehole samples in the Makululu settlement exceeded WHO safety limits for lead, exposing adults and children to persistent, daily toxic ingestion [25]. Furthermore, the failure to regulate manufacturing allows new sources of exposure to proliferate; Musa et al. (2025) identify that new decorative paints in Nigeria still contain dangerously high concentrations of lead, perpetuating the cycle of contamination in the built environment [24].

Even in developed nations, resource extraction creates “Sacrifice Zones” where marginalised populations bear disproportionate health burdens. In the mining town of Mount Isa, Australia, regulatory bodies and industry actors have historically obfuscated the link between mining emissions and community health. Forbes and Taylor (2015) argue that despite clear evidence of contamination, public health leadership has been reactive rather than proactive, resulting in Indigenous children suffering from significantly higher blood lead levels compared to their non-Indigenous peers [41]. Although representing conditions documented a decade ago, this case remains analytically relevant as a historically significant example of the regulatory failure pattern that continues to be documented in comparable LMIC contexts through more recent surveillance data. This disparity highlights a global failure to protect vulnerable communities from the externalised costs of resource extraction.

#### 4.4 Tradition vs. toxicology: the cultural blind spot

A distinct and often overlooked category of regulatory failure arises where deep-seated cultural practices intersect with toxicological hazards, effectively evading the industrial control frameworks designed for mass-manufactured goods. This “cultural blind spot” is

exemplified by the persistent use of lead-glazed ceramics (LGC) in Mexico, where traditional knowledge regarding safety measures has proven scientifically unfounded and hazardous. Rodríguez-Hernández et al. (2025) empirically debunked the efficacy of traditional “curing” methods such as boiling with vinegar or applying oil and heat, widely believed by artisans and consumers to seal pores and remove lead. Contrarily, their analysis revealed that these curing processes significantly increased the bioavailability of lead; the mean lead concentration on the surface of LGC pieces rose from 164,400 mg/kg before curing to 266,700 mg/kg afterwards [15]. Furthermore, under simulated cooking conditions, pots cured with oil and heat leached lead at concentrations as high as 378.18 mg/kg, exceeding the Mexican Normativity limit of 0.5 mg/kg by over 750 times, thereby transforming a cultural staple into a potent vector for dietary toxicity [15].

This phenomenon is not confined to the culinary arts or lower-income nations; it manifests as a “cosmetic paradox” in high-income economies as well. While often associated with traditional practices, lead exposure in the Middle East reveals a modern regulatory blind spot involving both cultural and commercial vectors. Ahmed et al. (2021) analysed cosmetic products in Saudi Arabian markets and identified significant lead contamination in traditional Kajal (Kohl), a popular eyeliner, reinforcing the risk of cultural vectors similar to South Asian Surma. However, the risk has evolved beyond tradition. Almukainzi et al. (2022), in a comprehensive quality assessment published in *Scientific Reports*, detected lead, arsenic, and cadmium in modern, commercially available brands of lipstick, eyeshadow, and foundation. Similarly, Alqahtani et al. (2024) found that “low-price” cosmetics imported into the Saudi market frequently contained lead residues, demonstrating that global supply chains effectively bypass national safety standards, introducing heavy metals into the daily routines of women and children in high-income economies [12–14].

This intersection of tradition and toxicity extends to religious practices, where regulatory oversight is frequently deferential or non-existent. Wadhwa and Arora (2022) identify significant lead risks in Sindur, a cosmetic powder applied to the hair parting of married Hindu women, which is traditionally coloured with lead tetroxide (red lead). Their analysis detected lead concentrations exceeding 300,000 mg/kg in samples, creating a direct pathway for dermal absorption and accidental ingestion; in one documented case, a child’s blood lead level reached toxic thresholds after lead-adulterated Sindur was mistakenly used as a food colouring agent [42]. The hazard also migrates across borders, bypassing the domestic regulations of developed nations through refugee resettlement and the transport of cultural heirlooms. Fellows et al. (2022) found that Afghan refugee children in the United States faced elevated blood lead levels linked to traditional aluminium pressure cookers brought from Afghanistan. These cookpots, often manufactured from scrap metal including car parts, contained lead concentrations exceeding 60,000 mg/kg and leached lead into food at levels surpassing the FDA’s interim reference level for children by over 650-fold [43]. These findings illustrate that cultural artefacts often function as Trojan horses for heavy metal exposure, permeating borders and regulatory shields that focus primarily on industrial emissions rather than traditional goods.

#### 4.5 A Model for success: the bangladesh turmeric intervention

Amidst the global landscape of regulatory failure and enforcement inertia, the recent intervention in the Bangladeshi turmeric sector demonstrates that targeted, regulatory

enforcement combined with rapid detection technology can effectively sever the link between commerce and toxicity. Forsyth et al. (2023) detail a multi-faceted campaign orchestrated between 2017 and 2021 to eliminate the addition of lead chromate, a neurotoxic pigment used to brighten roots, from the turmeric supply chain. Unlike the passive educational campaigns often seen in the informal sectors of other nations, the Bangladesh Food Safety Authority utilised mobile courts to impose substantial fines on wholesale traders and confiscate adulterated goods, utilising X-ray fluorescence (XRF) technology for immediate, on-site evidence generation [27].

The efficacy of this intervention was rapid and measurable. Forsyth et al. (2023) report that the proportion of market turmeric samples containing detectable lead plummeted from 47% pre-intervention in 2019 to 0% by 2021. Furthermore, direct evidence of lead chromate at polishing mills decreased from 30% to 0% over the same period. The human health implications were immediate; blood lead levels among turmeric mill workers dropped by a median of 30% within 16 months of the intervention [27]. This success stands in stark contrast to the persistent contamination observed in the Mexican ceramic and Indian e-waste sectors, suggesting that where regulatory authority exists to enforce food safety laws strictly rather than relying on voluntary compliance or consumer awareness, the eradication of established toxic practices is achievable in a short timeframe.

Collectively, these findings indicate that lead exposure persists primarily due to regulatory exemptions and enforcement asymmetry rather than scientific uncertainty (Table 1).

The regulatory contradictions and enforcement asymmetries documented above do not operate in isolation; they translate directly into the persistence and replenishment of environmental lead reservoirs through which human exposure ultimately occurs. The following section examines these exposure pathways in detail, tracing how governance failures at the policy level materialize as measurable contamination in soil, water, and the food chain.

## 5 Environmental reservoirs: the route of exposure

### 5.1 Atmospheric deposition and soil reservoirs

**Mechanism:** Airborne lead particulates emitted from industrial or aviation sources undergo atmospheric transport and deposition, settling into surface soil where they accumulate as a persistent environmental hazard that children ingest via “Hand-to-Mouth” behaviour [44, 45]. This resuspension of contaminated dust transforms soil into a chronic exposure pathway, particularly for paediatric populations residing near emission sources [10].

**Evidence:** In the United States, lead formulated in aviation fuel is emitted by piston-engine aircraft, resulting in significantly elevated blood lead levels (BLLs) for children living near airports; specifically, an increase in atmospheric lead of  $1 \mu\text{g}/\text{m}^3$  is associated with a  $4.05 \mu\text{g}/\text{dL}$  increase in child BLLs [10]. In contrast to aviation deposition, industrial recycling villages in Vietnam exhibit extreme soil contamination, where the median lead concentration on transportation routes reached  $10,000 \text{ mg}/\text{kg}$  (equivalent to 10,000 parts per million, ppm), with 99% of measurement points exceeding the residential standard of  $120 \text{ mg}/\text{kg}$  [45]. Similarly, in Owino Uhuru, Kenya, soil analysis near a former

lead smelter identified hotspots with lead concentrations reaching 5,824 mg/kg near the perimeter wall, presenting a critical risk for incidental oral ingestion [44].

## 5.2 Water systems: groundwater and infrastructure

**Mechanism:** Lead infiltration into water systems is governed by a combination of environmental hydrogeology and plumbing chemistry [46, 47]. In groundwater environments, lead mobilisation is frequently driven by Acid Mine Drainage (AMD), which occurs when legacy mining activities expose sulfidic minerals, such as galena and pyrite, to oxygen and water [46, 48]. This sulfide oxidation generates a highly acidic, sulfate-rich leachate that lowers the pH of surrounding water systems [48]. A reduction in pH alters metal speciation, inducing the desorption of lead from biological and physical surfaces and increasing the concentration and mobility of free lead ions [46, 49].

Within distribution infrastructure, lead enters drinking water primarily through the dissolution of lead pipes, solders, and brass fittings [47, 50]. Corrosion kinetics are initially mitigated by the formation of passivation scales, particularly highly insoluble Pb(IV) oxides like PbO<sub>2</sub>, which remain chemically stable under the high oxidation-reduction potential (ORP) maintained by free chlorine disinfectants [49, 51, 52]. Passivation scale degradation occurs when shifts in water chemistry such as pH fluctuations, natural organic matter intrusion, or switching disinfectants to monochloramine reduce the ORP, triggering the reductive dissolution of protective PbO<sub>2</sub> into highly soluble Pb(II) phases [51, 52]. Furthermore, infrastructure containing bimetallic junctions, such as lead-copper connections, undergoes galvanic corrosion [49, 53]. In these macrocells, lead typically acts as the anode relative to the more noble copper or brass, leading to accelerated oxidation of the lead, the formation of an acidic microenvironment, and the migration of sulfate and chloride anions toward the anodic joint [49]. This galvanic interaction deposits thickened, structurally unstable lead rust and scales at the junction [53]. Subsequent high flow rates apply hydraulic shear stress that dislodges these accumulated deposits, resulting in the erratic release of particulate lead into the water stream [47, 52, 53]. These mechanistic pathways are corroborated by field and experimental evidence across geographically distinct settings.

**Evidence:** Historical mining operations profoundly degrade regional hydrogeology, as demonstrated by groundwater contamination in the Makululu township of Kabwe, Zambia [25]. Infiltration from local lead and zinc mine tailings resulted in severe contamination, with 100% of tested hand pump boreholes yielding a median lead concentration of 150 µg/L, and 77% of shallow wells yielding a median of 60 µg/L and reaching a maximum of 380 µg/L. These values drastically exceed both the World Health Organization (WHO) maximum acceptable limit of 10 µg/L and the United States Environmental Protection Agency (US EPA) action limit of 15 µg/L [25]. Deep boreholes reaching 50 to 100 m exhibited significantly higher lead concentrations than 10-meter shallow wells, indicating that deeper aquifers experience greater interaction with mineral-rich geological zones affected by mining leachate [25].

In addition to environmental groundwater sources, municipal infrastructural components present distinct contamination pathways. A sampling of 261 community water systems across rural Ghana, Mali, and Niger demonstrated that 9% of drinking water samples surpassed the WHO threshold of 10 µg/L [50]. Analysis of the plumbing components isolated lead-containing brass fixtures as the primary source of this leaching,

underscoring the risk posed by specific plumbing hardware even in the absence of pure lead pipes [50].

The impact of plumbing architecture is further quantified by pilot studies simulating partial lead service line replacements, which create active galvanic lead-copper junctions. Under medium and high flow rates, service lines configured with 50% lead piping placed downstream of copper released an average of 3.5 times more lead than systems composed entirely of 100% lead pipe [53]. During these higher velocity flow events, at least 85% of the total lead measured was in particulate form, having been physically sheared from the massive galvanic deposits at the junction [53]. These galvanic particulate spikes frequently exceeded regulatory action thresholds, demonstrating that conventional low-flow sampling methodologies fundamentally underestimate lead exposure in dynamically flowing water systems [53].

### 5.3 The food chain and consumer vectors

**Bioaccumulation:** Plants growing in contaminated environments absorb lead through their root systems, bioaccumulating the toxicant in edible tissues. In the Dong Mai recycling area of Vietnam, cultivated rice grains (*O. sativa*) contained up to 2.83 mg/kg dry weight (DW) of lead, while the native hyperaccumulator *Hymenachne acutigluma* (Steud.) reached shoot concentrations of 1,160 mg/kg DW [54].

**Adulteration & Leaching:** Direct oral exposure also occurs through food adulteration and the use of unsafe cookware. In Bangladesh, lead chromate pigment containing 2–10% lead by weight is added to turmeric roots to enhance colour, resulting in turmeric powder lead concentrations as high as 1,152 mg/kg [8]. Additionally, traditional Mexican ceramics cured with vegetable oil and heat were found to leach 378.18 mg/L of lead into acidic food simulants, vastly exceeding the Mexican Normativity limit of 0.5 mg/L by over 750-fold [15]. Furthermore, imported aluminium cookware used by Afghan refugees released dangerous quantities of lead, with pressure cookers delivering estimated daily doses of up to 1,940 µg/day [43].

## 6 Internal dose and surveillance reality: blood lead levels as the bridge

### 6.1 The ‘gold standard’ fallacy: blood vs. bone

**The Half-Life Problem:** Current toxicological surveillance remains dangerously limited by a reliance on Blood Lead Levels (BLL) as the primary clinical metric, despite evidence that BLL is pharmacokinetically restricted to reflecting recent acute exposure, with a half-life of approximately 4–5 weeks [55, 56]. This short-term window fails to capture the true body burden, as lead sequestered in the skeletal system persists with a half-life spanning 20 to 25 years, meaning that standard testing protocols frequently underestimate the chronic toxicity risk in populations with historical rather than concurrent exposure [56].

**The Silent Reservoir:** The physiological reality of lead kinetics is that 60% to 90% of the total body burden in children and up to 90–99% in adults is sequestered within the skeletal system rather than the circulation [55, 57]. This skeletal storage acts as a ‘silent reservoir’ that is not metabolically inert; stored lead can be endogenously remobilised back into the bloodstream during periods of physiological stress or therapeutic chelation, perpetuating internal exposure long after external sources are removed [57].

## 6.2 Systemic toxicity: bone, teeth, and metabolism

**Skeletal Damage:** The cumulative storage of lead in bone drives mechanistic disruption of calcium homeostasis by inhibiting the renal enzyme 1-alpha-hydroxylase, resulting in significantly suppressed serum vitamin D3 levels and elevated markers of bone resorption such as type I collagen cross-linked C-telopeptide (CTX-1) [58]. This metabolic interference manifests structurally as reduced bone mineral density, specifically observed as significantly lower T-scores and Z-scores in the lumbar vertebrae of lead-exposed patients compared to healthy controls [59].

**Oral Health:** The systemic toxicity of lead extends beyond the skeleton to the oral cavity, where recent meta-analytic data confirm a significant positive association between lead exposure and periodontitis (adjusted OR = 1.85), driven by immune dysregulation and oxidative stress [60]. These findings further highlight the capacity of calcified tissues to accumulate heavy metals, with studies utilising tooth deposition as a validated biomarker for historical exposure, reinforcing that the oral cavity is a critical site of heavy metal pathology [60].

## 6.3 The epigenetic and cognitive long-tail

**Epigenetics:** Advanced molecular epidemiology has exposed the failure of BLL to predict neurodegeneration; utilising novel DNA methylation biomarkers to estimate cumulative exposure, researchers demonstrated that higher estimated tibia and patella lead levels were significantly associated with lower Montreal Cognitive Assessment (MoCA) scores [61]. Crucially, this epigenetic analysis revealed that while cumulative bone lead estimates predicted cognitive decline, particularly in women and APOE4 carriers, recent DNA methylation-estimated blood lead levels showed no such association, proving that long-term storage is the primary driver of neural decay [61].

**The LMIC Gap:** This surveillance gap is most profound in low- and middle-income countries (LMICs) like South Africa, where elderly populations residing near historical gold mining operations exhibit high rates of probable early dementia (55%) and deficits in executive function that correlate with cumulative bone lead levels [62]. Standard blood tests in these regions fail to capture the neurotoxic legacy of historical exposure, leaving vulnerable populations with “silent” skeletal burdens that drive cognitive impairment, specifically in domains such as spatial working memory and impulse control [62].

## 7 Neurobiological mechanisms: the molecular price

Mechanistic evidence is derived from a combination of in vitro neuronal models, rodent developmental exposure studies, and human epidemiological associations. While exact molecular targets vary across models and exposure windows, convergent evidence supports synaptic disruption and mitochondrial injury as core pathways.

### 7.1 The ‘Trojan horse’: ionic mimicry and BBB transport

The neurotoxicity of lead ( $\text{Pb}^{2+}$ ) is fundamentally driven by its capacity to function as a molecular “Trojan Horse,” effectively mimicking the essential divalent cation calcium ( $\text{Ca}^{2+}$ ) due to their homologous bio-physicochemical properties and similar ionic radii (1.19 Å for  $\text{Pb}^{2+}$  vs. 1.00 Å for  $\text{Ca}^{2+}$ ), which allows  $\text{Pb}^{2+}$  to competitively displace calcium at critical regulatory sites [29]. This ionic mimicry facilitates the translocation of  $\text{Pb}^{2+}$  across the Blood-Brain Barrier (BBB), where it hijacks endogenous calcium transport



systems, likely utilising calcium transport pathways and voltage-gated calcium channels to accumulate within the brain parenchyma [29, 30].

### 7.2 NMDA receptor blockade and synaptic plasticity impairment

Upon entry into the synapse,  $Pb^{2+}$  acts as a potent non-competitive antagonist by binding with high affinity to the zinc ( $Zn^{2+}$ ) allosteric regulatory site located on the GluN2A subunits of the N-methyl-D-aspartate (NMDA) receptor, thereby suppressing channel function in a voltage-independent manner [32]. This receptor blockade critically impedes the influx of extracellular calcium, consequently inhibiting the phosphorylation and activation of calcium/calmodulin-dependent protein kinase II (CaMKII) and the cAMP response element-binding protein (CREB), both of which are obligatory for the gene transcription required for synaptic plasticity [33]. Ultimately, the disruption of this signalling cascade abolishes the induction of Long-Term Potentiation (LTP), the cellular substrate of learning and memory, and significantly reduces the field excitatory postsynaptic potential (fEPSP), resulting in profound synaptic dysfunction [34].

### 7.3 Mitochondrial dysfunction, apoptosis, and long-term neurodegenerative risk

At the organellar level,  $Pb^{2+}$  exploits mitochondrial calcium transport pathways to gain entry into the mitochondrial matrix, with experimental evidence demonstrating that modulation of the mitochondrial calcium uniporter (MCU) directly alters  $Pb^{2+}$ -induced cytotoxicity [63]. Once internalised,  $Pb^{2+}$  precipitates severe Endoplasmic Reticulum (ER) stress that disrupts mitochondrial dynamics and upregulates the pro-fission GTPase Dynamin-related protein 1 (DRP1), driving the organelle toward excessive fragmentation and structural collapse [37]. This bioenergetic failure and concurrent oxidative stress converge to induce the opening of the Mitochondrial Permeability Transition Pore (mPTP), triggering the release of cytochrome c into the cytosol and the sequential activation of caspase-9 and caspase-3 cascades that execute apoptotic cell death [38]. Over extended periods of exposure, the sustained calcium dysregulation and pro-oxidant environment generated by these mitochondrial injuries create a pro-amyloidogenic cellular state that promotes the aberrant processing and accumulation of Amyloid-Beta peptide ( $A\beta_{42}$ ), providing a plausible mechanistic link between early-life lead exposure and elevated risk of Alzheimer's disease pathogenesis in later life [4].

Taken together, these interconnected pathways explain how  $Pb^{2+}$  exposure translates into synaptic failure, mitochondrial collapse, and long-term neurodegenerative risk; the key molecular events are summarised in Table 2.

## 8 The global burden: economic and health consequences

### 8.1 Economic burden of persistent lead exposure

The Global Cost: The continued presence of lead in the environment is not merely a health issue; it is a substantial and preventable economic burden. Contemporary childhood lead exposure now costs the global economy an estimated \$3.4 trillion USD annually, a staggering sum that equates to approximately 2% of the global gross domestic product (GDP) [2]. This substantial financial loss stems primarily from the cumulative cognitive damage inflicted upon children, which permanently stunts their future productivity and lifetime earning potential [2].

**The Inequality Gap:** This burden is not shared equally; it falls disproportionately on those least able to afford it. Low- and middle-income countries (LMICs) suffer the most severe economic consequences, with regions such as Africa and Asia experiencing relative lifetime earning losses exceeding 10%, nearly 9% of GDP in effective terms, compared to far lower losses in high-income nations [2]. This economic drain effectively neutralises development aid, as the prevalence of elevated blood lead levels remains critically high in these lower-income regions, where regulatory gaps allow contaminated consumer products and industrial emissions to proliferate unchecked [7].

## 8.2 The neurological impairment and the cycle of poverty

**IQ Loss:** Lead acts as a potent neurotoxicant that causes permanent, irreversible structural damage to the developing brain, with no established safe threshold of exposure [64]. A global health impact modelling analysis estimated that children younger than five years collectively lost 765 million IQ points (95% CI: 443–1,098 million) due to lead exposure in 2019 alone, of which 95.3% occurred in low- and middle-income countries (LMICs) [64]. At the individual level, children in LMICs experienced an average loss of 5.9 IQ points during their first five years of life, translating to an estimated lifetime income reduction of nearly 12% per affected child [64]. The aggregate economic cost of this IQ loss was estimated at US\$1.4 trillion, equivalent to 1.6% of global GDP in 2019, rising to 8.3% of GDP in low-income countries, where the burden is most severe [64]. This neurological impairment directly reduces educational attainment and lifetime earning potential, perpetuating intergenerational cycles of poverty in the populations most heavily exposed [64, 65].

**The Poverty Trap:** This widespread cognitive impairment functions as a trap, locking nations into a cycle of poverty. In regions like Sub-Saharan Africa, where environmental risks such as lead exposure are endemic and syndemic with other stressors, the irreversible neurodevelopmental harm inflicted on children undermines the human capital necessary for economic development [65]. The failure to address these toxicant exposures perpetuates a system where vulnerable populations are unable to escape economic stagnation due to the silent, irreversible loss of their cognitive potential [65].

## 8.3 Beyond the brain: systemic toxicity and fertility

**Reproductive Health:** The toxicity of lead extends well beyond the nervous system, striking at the very capacity for population renewal. Lead exposure is a proven reproductive toxicant that significantly reduces fertility rates and increases the incidence of foetal death and infant mortality [66]. Furthermore, exposure during pregnancy is associated with severe adverse birth outcomes, including miscarriage, preterm birth, and low birth weight, as lead stored in maternal bone is mobilised and transferred to the foetus [66, 67].

**Organ Damage:** Once ingested or inhaled, lead distributes throughout the body, accumulating in soft tissues such as the kidneys and liver [30]. This bioaccumulation causes chronic organ dysfunction and induces systemic oxidative stress, disrupting essential cellular processes and antioxidant defence systems far beyond the neural networks [3, 30]. The resulting oxidative damage contributes to a host of chronic conditions, further exacerbating the public health burden on already strained healthcare systems [3].

#### 8.4 The cost of inaction

Section conclusion: The evidence indicates that the cost of inaction is unsustainable. Implementing primary prevention strategies and remediation to remove lead from the environment is far more cost-effective than the exorbitant expense of treating the life-long health consequences and absorbing the economic productivity losses associated with exposure [67]. Universal screening and immediate risk reduction are not just medical imperatives but economic necessities to prevent the continued erosion of human potential [67].

### 9 Strategic recommendations: a roadmap for elimination

#### 9.1 Global trade harmonisation: closing the 'export loophole'

The Problem: High-income nations ban toxic chemicals at home but export them to the developing world. Evidence confirms that countries such as the United States and Canada maintain stringent domestic standards for lead in paint (e.g., 90 mg/kg) yet continue to export significant quantities of lead chromate pigments to low- and middle-income countries (LMICs) with weaker regulatory capacity [6].

Proposed Solution: The EU and the USA should adopt mirror clauses in trade policy, such that if a chemical is banned domestically, its export is also prohibited. High-income countries should prioritise ending the export of hazardous lead chromate to LMICs to avoid sustaining industries that contribute to population-level poisoning. Ultimately, all countries should move toward a complete ban on the manufacture and use of lead chromate, including for export [6].

Supply Chain Audits: We recommend mandatory heavy metal testing for consumer goods (spices/toys) imported from LMICs, as 18% of goods currently fail tests. A rapid market screening of over 5,000 products across 25 LMICs revealed that 18% of samples exceeded reference levels for lead, including high rates in metal foodware, ceramics, and paints, necessitating rigorous supply chain analysis [7].

#### 9.2 The 'Bangladesh model' for food safety

This intervention provides a scalable model for replication, combining rapid market screening with enforceable regulatory action (e.g., fines and penalties). In Bangladesh, a multi-component strategy involving media campaigns, business engagement, and government enforcement supported by rapid lead-detection technology reduced the proportion of market turmeric samples with detectable lead from 47% (pre-intervention) to 0% (post-intervention) [27].

#### 9.3 Formalising the informal sector: a community approach

The Problem: Punitive bans on battery recycling drive pollution underground. Complex regulatory frameworks and high tax burdens discourage engagement with the government, leading informal actors to operate underground using counterfeit licences and hazardous processing techniques to avoid financial losses [17].

Proposed Solution: We recommend 'Green Recycling Zones' with incentives. Crucially, emphasise 'Community-Based Participatory Research' (CBPR). Explain that interventions consistently fail unless the community is involved in designing them (e.g., in cultural pottery or recycling) [68]. CBPR programmes empower disenfranchised communities by including them as equal partners in the research process, which is essential

for ensuring that educational interventions regarding lead risks, such as those involving soil or cultural products, are culturally and linguistically appropriate [68].

#### 9.4 From 'secondary' to 'primary' prevention

**The Shift:** This review argue that the world relies too much on 'Secondary Prevention' (screening children after they are poisoned). Current practices often rely on identifying individuals with the disease who lack symptoms to prevent severe consequences, rather than preventing the exposure initially [67].

**Proposed Solution:** We must shift to 'Primary Prevention' (screening housing, water, and soil before a child is born/exposed). This is the only way to stop the 'Cycle of Toxicity'. Nurses and healthcare providers must adopt a public health framework that focuses on educating individuals and identifying sources of lead in the home environment, such as water, paint, and soil, before pregnancy or childhood exposure [67].

**Funding:** We recommend implementing a 'Polluter-Pays' tax on aviation/mining to fund soil remediation. To recover lead-contaminated urban environments and protect public health, mandatory testing and the funding of soil emplacement (covering contaminated soil) are necessary and feasible next steps, shifting accountability to toxicant producers under a precautionary framework [1].

### 10 Knowledge gaps and future directions

Despite substantial advances in understanding the sources, mechanisms, and consequences of lead neurotoxicity, critical gaps persist across four interconnected domains that warrant prioritised attention.

First, representative blood lead level surveillance in LMICs remains the most urgent data gap. Current biomonitoring infrastructure is severely uneven. Sub-Saharan Africa outside South Africa, and much of South and Southeast Asia beyond India, lack nationally representative, periodically administered blood lead level measurement programmes. Existing estimates from these regions rely on convenience samples from urban populations, hospital-based studies, or occupationally exposed cohorts, systematically excluding rural communities and children in informal settlements where exposure burdens are likely highest. Harmonising blood lead surveillance with existing household survey platforms such as the Demographic Health Surveys or Multiple Indicator Cluster Surveys would provide cost-effective, nationally representative data at scale and enable meaningful cross-country comparisons for the first time.

Second, the field requires a methodological transition beyond blood lead levels toward cumulative lifetime exposure assessment. Blood lead level reflects only recent exposure and rapidly declines following source removal, rendering it inadequate as a marker of the cumulative neurological burden accumulated over years of chronic low-level exposure. K-XRF bone lead measurement, which directly quantifies the skeletal lead reservoir accumulated over a lifetime, offers a validated alternative that is increasingly feasible in field settings. Similarly, emerging epigenetic biomarkers, particularly lead-associated DNA methylation signatures at loci including BDNF and the LINE-1 repetitive element, offer the potential for non-invasive retrospective exposure reconstruction. Prospective cohort studies incorporating both bone lead and methylation biomarkers are urgently needed in LMIC populations to accurately characterise the true neurotoxicological burden.

Third, source attribution methodology requires systematic strengthening through isotopic fingerprinting and receptor modelling. In most affected communities, the relative contributions of legacy soil contamination, current consumer product exposure, occupational pathways, and food chain bioaccumulation to individual blood lead levels remain poorly quantified. Without robust source attribution, intervention resources cannot be rationally prioritized. Isotopic lead fingerprinting combined with receptor modelling has been successfully applied in high-income country settings to discriminate between these sources, but remains almost entirely absent from LMIC research programmes. Establishing standardised source attribution protocols adapted for low-resource settings would directly inform which regulatory and remediation interventions are most likely to reduce exposure in specific geographic contexts.

Fourth, the intervention evidence base contains critical prospective gaps that limit evidence-based policy translation. Several policy recommendations that appear throughout the lead neurotoxicity literature including informal sector formalisation, green recycling zone development, and community-based participatory remediation currently rest on plausible theoretical frameworks and cross-sectional associations rather than prospective evaluation evidence. Rigorous prospective studies are needed in three specific areas: enforcement evaluation of informal sector formalization programmes, measuring blood lead level outcomes in exposed workers before and after regulatory transition; cost-effectiveness analysis of soil remediation approaches compared with population-level screening and chelation, disaggregated by exposure context; and randomised or quasi-experimental evaluation of food adulteration enforcement interventions, using the Bangladesh turmeric chromate elimination programme as the benchmark model for replication and adaptation.

Addressing these four gaps surveillance harmonization, cumulative exposure assessment, source attribution, and prospective intervention evaluation represents the minimum research agenda required to translate the scientific consensus on lead neurotoxicity into measurable reductions in population exposure, particularly in the LMICs that bear a disproportionate share of the global burden.

## 11 Limitations

This review is limited by its narrative synthesis design and therefore does not provide a PRISMA-based systematic search, meta-analysis, or formal risk-of-bias scoring across included studies. Additionally, exposure evidence across regions varies in measurement methods, surveillance intensity, and population representativeness, which may introduce heterogeneity in reported blood lead levels and environmental contamination estimates. Despite these limitations, the objective of this review is to provide an integrated mechanistic and policy framework that explains why lead exposure persists globally and highlights actionable leverage points for elimination-focused interventions.

## 12 Conclusion: from chemical safety to political will

This review demonstrates that the persistence of lead neurotoxicity in the 21st century is not a failure of scientific understanding, but a failure of governance. The biological mechanisms are well-established: lead acts as a molecular mimic, compromising the blood-brain barrier to disrupt NMDA receptors, impair synaptic plasticity, and promote mitochondrial dysfunction and apoptosis. The sources are equally well-identified: from

the “regulatory asymmetry” of aviation fuel exemptions and toxic exports in the Global North, to the “survival economics” of informal battery recycling and adulterated spices in the Global South.

Current regulatory frameworks are insufficient. Developed nations must close the export loopholes that allow them to externalise toxic risks to developing economies. Simultaneously, low- and middle-income nations must move beyond “paper bans” to address the economic drivers of the informal sector, utilising successful models like the Bangladesh turmeric intervention that combine strict enforcement with public awareness. As long as the economic cost of compliance exceeds the value placed on human health in policy calculations, lead will remain a policy-permitted neurotoxic exposure. The \$3.4 trillion annual loss is not merely a statistic; it represents a permanent tax on the cognitive potential of future generations. Eliminating lead exposure is directly aligned with the United Nations Sustainable Development Goals (SDGs), particularly SDG 3 (Good Health and Well-being), SDG 6 (Clean Water and Sanitation), and SDG 12 (Responsible Consumption and Production), underscoring that lead prevention is not only a toxicological imperative but a global development priority. The era of merely “managing” lead exposure must end; a strategic transition from exposure management to primary prevention and elimination represents the most evidence-aligned path forward.

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The authors confirm that artificial intelligence-based tools were used solely for language editing, including grammar and clarity improvements. The authors remain fully responsible for the content of this manuscript.

#### **Author contributions**

Naveen Kumar Rajasekaran: Writing-original draft, Writing-review and editing, Formal analysis, Conceptualization, Data curation, Visualization, and Project administration. Chetan Ashok: Writing-review and editing, Formal analysis, Conceptualization, Data curation, and Visualization. Srikanth Jeyabalan: Writing-review and editing, Supervision, Formal analysis, Conceptualization, Visualization, and Project administration. Gayathri Veeraraghavan: Writing-review and editing, Formal analysis, and Conceptualization. Vetrivelan Subramanian: Writing-review and editing, Formal analysis, and Conceptualization. Devaraja Dravida Pandian S: Writing-review and editing, Formal analysis, and Conceptualization. Ling Shing Wong: Writing-review and editing, Formal analysis, and Conceptualization. Mahendran Sekar: Writing-review and editing, Formal analysis, and Conceptualization. Ravindran Jaganathan: Writing-review and editing, Formal analysis, and Conceptualization.

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## References

1. Mielke H, Egendorf S. Getting the Lead Out: A Career-Long Perspective on Leaded Gasoline, Dust, Soil, and Proactive Pediatric Exposure Prevention. *Med Res Arch*. 2023;11. <https://doi.org/10.18103/mra.v11i5.3813>.
2. Chen M, Gazze L, DiTraglia FJ, Das R, Nriagu J, Erel Y, et al. Environmental lead risk in the 21st century. *Commun Earth Environ*. 2025;6:776. <https://doi.org/10.1038/s43247-025-02735-x>.
3. Ahmed G, Rahaman MS, Perez E, Khan KM. Associations of Environmental Exposure to Arsenic, Manganese, Lead, and Cadmium with Alzheimer's Disease: A Review of Recent Evidence from Mechanistic Studies. *J Xenobiotics*. 2025;15. <https://doi.org/10.3390/jox15020047>.
4. Xie J, Wu S, Szadowski H, Min S, Yang Y, Bowman AB, et al. Developmental Pb exposure increases AD risk via altered intracellular Ca<sup>2+</sup> + homeostasis in hiPSC-derived cortical neurons. *J Biol Chem*. 2023;299. <https://doi.org/10.1016/j.jbc.2023.105023>.
5. Poudel K, Ketema RM, Ngo HTT, Ikeda A, Minatoya M. E-waste in Vietnam: a narrative review of environmental contaminants and potential health risks. *Rev Environ Health*. 2024;39:391–406. <https://doi.org/10.1515/reveh-2022-0233>.
6. Todd R, Crawford L, Bonfield R. Toxic Lead Paint Pigment Exports from Rich to Poor Countries. Policy Pap. 2025.
7. Sargsyan A, Nash E, Binkhorst G, Forsyth JE, Jones B, Sanchez Ibarra G, et al. Rapid Market Screening to assess lead concentrations in consumer products across 25 low- and middle-income countries. *Sci Rep*. 2024;14:9713. <https://doi.org/10.1038/s41598-024-59519-0>.
8. Forsyth JE, Nurunnahar S, Islam SS, Baker M, Yeasmin D, Islam MS, et al. Turmeric means yellow in Bengali: Lead chromate pigments added to turmeric threaten public health across Bangladesh. *Environ Res*. 2019;179:108722. <https://doi.org/10.1016/j.envres.2019.108722>.
9. Soale A-N, Callender R, Guignet D, Shadbegian R, Miranda ML. Association between Residential Distance to Airport and Blood Lead Levels in Children under 6 Living in North Carolina, 1992–2015. *Environ Health Perspect*. 2024;132:087701. <https://doi.org/10.1289/EHP14362>.
10. Zahran S, Keyes C, Lanphear B. Leaded aviation gasoline exposure risk and child blood lead levels. *PNAS Nexus*. 2023;2:pgac285. <https://doi.org/10.1093/pnasnexus/pgac285>.
11. Prakash Sahoo J, Chandra Samal K. Adulteration in Indian spices: An alarming concern and a silent health hazard. *Int J Adulteration*. 2024;8:1–9. <https://doi.org/10.54905/disssi.v8i9.e4ijad3043>.
12. Ahmed AY, Asada A, Hamza IAA. Cobalt and lead concentrations in cosmetic products sold at local markets in Saudi Arabia. *Toxicol Rep*. 2021; 8:1693–1698. <https://doi.org/10.1016/j.toxrep.2021.09.004>.
13. Almukainzi M, Alotaibi L, Abdulwahab A, Albukhary N, El Mahdy AM. Quality and safety investigation of commonly used topical cosmetic preparations. *Sci Rep*. 2022;12:18299. <https://doi.org/10.1038/s41598-022-21771-7>.
14. Alqahtani AM, Mojally M, Sayqal A, Ainousah BE, Alqmash A, Alzahrani S, et al. Determination of lead and cadmium concentration in cosmetic products in the Saudi market. *J Umm Al-Qura Univ Appl Sci*. 2024;10:146–55. <https://doi.org/10.1007/s43994-023-00088-9>.
15. Rodríguez-Hernández MF, Betanzos-Robledo L, Mariscal-Moreno RM, Valverde-Arámbula FA, Chuck-Hernández C, Peralta-Delgado N, et al. Evaluation of Three Traditional Curing Methods Applied in Mexican Lead-Glazed Ceramics: Detection, Concentration, and Leaching of Lead to Food. Processes. 2025;13. <https://doi.org/10.3390/pr13092766>.
16. Peralta N, Cantoral A, Téllez-Rojo MM, Trejo-Valdivia B, Estrada-Sánchez D, Richardson -LV, et al. Lead Levels in a Potters Population and Its Association With the Use of Different Glazes: Cross-Sectional Evaluation of the Approved Pottery Program. *Front Toxicol*. 2022;4. <https://doi.org/10.3389/ftox.2022.799633>.
17. Heinrich V, Bansal H, Fröhling M. Electronic waste management in Northern India: a regional case study of Chandigarh. *J Mater Cycles Waste Manag*. 2025;27:1944–61. <https://doi.org/10.1007/s10163-025-02194-8>.
18. Sharma L, Pandey S. Recovery of resources from end-of-life passenger cars in the informal sector in India. *Sustain Prod Consum*. 2020;24:1–11. <https://doi.org/10.1016/j.spc.2020.06.005>.
19. Raj K, Das AP. Lead pollution: Impact on environment and human health and approach for a sustainable solution. *Environ Chem Ecotoxicol*. 2023;5:79–85. <https://doi.org/10.1016/j.enceco.2023.02.001>.
20. Dang TD, Tran TA. Rural Industrialization and Environmental Governance Challenges in the Red River Delta, Vietnam. *J Environ Dev*. 2020;29:420–48. <https://doi.org/10.1177/1070496520942564>.
21. Daniell WE, Van Tung L, Wallace RM, Havens DJ, Karr CJ, Bich Diep N, et al. Childhood Lead Exposure from Battery Recycling in Vietnam. *BioMed Res Int*. 2015;2015:193715. <https://doi.org/10.1155/2015/193715>.
22. Ericson B, Duong TT, Keith J, Nguyen TC, Havens D, Daniell W, et al. Improving human health outcomes with a low-cost intervention to reduce exposures from lead acid battery recycling: Dong Mai, Vietnam. *Environ Res*. 2018;161:181–7. <https://doi.org/10.1016/j.envres.2017.10.042>.
23. Tirima S, Bartrem C, von Lindern I, von Braun M, Lind D, Anka SM, et al. Environmental Remediation to Address Childhood Lead Poisoning Epidemic due to Artisanal Gold Mining in Zamfara, Nigeria. *Environ Health Perspect*. 2016;124:1471–8. <https://doi.org/10.1289/ehp.1510145>.
24. Musa A, Hussaini A, Ali AS, Bello I, Sani S, Musa MD, et al. Lead Paint Hazards in Nigeria: Public Health and Environmental Impacts. *Technoscience J Community Dev Afr*. 2025;4:149–57. <https://doi.org/10.5281/zenodo.18108566>.
25. Siame T, Muzandu K, Mulenga KK, Dzombe CB. Lead-contaminated groundwater exposes residents to health risks in Makululu, Zambia. *J Water Health*. 2025;23:615–29. <https://doi.org/10.2166/wh.2025.343>.
26. Yabe J, Nakayama SMM, Nakata H, Toyomaki H, Yohannes YB, Muzandu K, et al. Current trends of blood lead levels, distribution patterns and exposure variations among household members in Kabwe, Zambia. *Chemosphere*. 2020;243:125412. <https://doi.org/10.1016/j.chemosphere.2019.125412>.
27. Forsyth JE, Baker M, Nurunnahar S, Islam S, Islam MS, Islam T, et al. Food safety policy enforcement and associated actions reduce lead chromate adulteration in turmeric across Bangladesh. *Environ Res*. 2023;232:116328. <https://doi.org/10.1016/j.envres.2023.116328>.
28. Forsyth JE, Mistree D, Nash E, Angrish M, Luby SP. Evidence of turmeric adulteration with lead chromate across South Asia. *Sci Total Environ*. 2024;949:175003. <https://doi.org/10.1016/j.scitotenv.2024.175003>.
29. Generalova A, Davidova S, Satchanska G, Generalova A, Davidova S, Satchanska G. The Mechanisms of Lead Toxicity in Living Organisms. *J Xenobiotics*. 2025;15. <https://doi.org/10.3390/jox15050146>.
30. Collin MS, Venkatraman SK, Vijayakumar N, Kanimozhi V, Arbaaz SM, Stacey RGS, et al. Bioaccumulation of lead (Pb) and its effects on human: A review. *J Hazard Mater Adv*. 2022;7:100094. <https://doi.org/10.1016/j.hazadv.2022.100094>.

31. Wu D, Chen Q, Chen X, Han F, Chen Z, Wang Y. The blood–brain barrier: Structure, regulation and drug delivery. *Signal Transduct Target Ther*. 2023;8:217. <https://doi.org/10.1038/s41392-023-01481-w>.
32. Tamagno WA, Freeman JL. Glutamate-Mediated Neural Alterations in Lead Exposure: Mechanisms, Pathways, and Phenotypes. *Toxics*. 2025;13. <https://doi.org/10.3390/toxics13070519>.
33. Chandravanshi LP, Agrawal P, Darwish HW, Trigun SK. Impairments of Spatial Memory and N-methyl-D-aspartate Receptors and Their Postsynaptic Signaling Molecules in the Hippocampus of Developing Rats Induced by As, Pb, and Mn Mixture Exposure. *Brain Sci*. 2023;13. <https://doi.org/10.3390/brainsci13121715>.
34. Guan R, Wang T, Dong X, Du K, Li J, Zhao F, et al. Effects of co-exposure to lead and manganese on learning and memory deficits. *J Environ Sci*. 2022;121:65–76. <https://doi.org/10.1016/j.jes.2021.09.012>.
35. Mohanraj N, Joshi NS, Poulouse R, Patil RR, Santhoshkumar R, Kumar A, et al. A proteomic study to unveil lead toxicity-induced memory impairments invoked by synaptic dysregulation. *Toxicol Rep*. 2022;9:1501–13. <https://doi.org/10.1016/j.toxrep.2022.07.002>.
36. Wei L, He H, Yang S, Shi Q, Wang X, Huang L, et al. Synergistic suppression of BDNF via epigenetic mechanism deteriorating learning and memory impairment caused by Mn and Pb co-exposure. *Ecotoxicol Environ Saf*. 2024;277:116365. <https://doi.org/10.1016/j.ecoenv.2024.116365>.
37. Zhang J, Su P, Xue C, Wang D, Zhao F, Shen X, et al. Lead Disrupts Mitochondrial Morphology and Function through Induction of ER Stress in Model of Neurotoxicity. *Int J Mol Sci*. 2022;23. <https://doi.org/10.3390/ijms231911435>.
38. Korotkov SM. Mitochondrial Oxidative Stress Is the General Reason for Apoptosis Induced by Different-Valence Heavy Metals in Cells and Mitochondria. *Int J Mol Sci*. 2023;24. <https://doi.org/10.3390/ijms241914459>.
39. Ayyalasomayajula N, Bandaru LJM, Chetty CS, Dixit PK, Challa S. Mitochondria-Mediated Moderation of Apoptosis by EGCG in Cytotoxic Neuronal Cells Induced by Lead (Pb) and Amyloid Peptides. *Biol Trace Elem Res*. 2022;200:3582–93. <https://doi.org/10.1007/s12011-021-02959-w>.
40. Jones MS, Allen AB. FAA EAGLE Avgas Transition: Considerations for Impacts on Alaskan Supply Chains. Report. Institute of Social and Economic Research; 2025.
41. Forbes M, Taylor MP. A review of environmental lead exposure and management in Mount Isa, Queensland. *Rev Environ Health*. 2015;30:183–9. <https://doi.org/10.1515/reveh-2015-0011>.
42. Wadhwa DD, Arora DrG. Identification of Lead from Sindur Samples. *Int J Innov Res Eng Manag*. 2022;357–60. <https://doi.org/10.55524/ijirem.2022.9.1.73>.
43. Fellows KM, Samy S, Rodriguez Y, Whittaker SG. Investigating aluminum cookpots as a source of lead exposure in Afghan refugee children resettled in the United States. *J Expo Sci Environ Epidemiol*. 2022;32:451–60. <https://doi.org/10.1038/s41370-022-00431-y>.
44. Ericson B, Otieno VO, Nganga C, St. Fort J, Taylor MP. Assessment of the Presence of Soil Lead Contamination Near a Former Lead Smelter in Mombasa, Kenya. *J Health Pollut*. 2019;9:190307. <https://doi.org/10.5696/2156-9614-9.21.190307>.
45. Fujimori T, Eguchi A, Agusa T, Tue NM, Suzuki G, Takahashi S, et al. Lead contamination in surface soil on roads from used lead–acid battery recycling in Dong Mai, Northern Vietnam. *J Mater Cycles Waste Manag*. 2016;18:599–607. <https://doi.org/10.1007/s10163-016-0527-7>.
46. Byrne P, Wood PJ, Reid I. The Impairment of River Systems by Metal Mine Contamination: A Review Including Remediation Options. *Crit Rev Environ Sci Technol*. 2012;42:2017–77. <https://doi.org/10.1080/10643389.2011.574103>.
47. Jarvis P, Fawell J. Lead in drinking water – An ongoing public health concern? *Curr Opin Environ Sci Health*. 2021;20:100239. <https://doi.org/10.1016/j.coesh.2021.100239>.
48. Barroso A, Valente TM, Antunes IMHR, Cerqueira A, Gomes P, Henriques R, et al. Integrated assessment of environmental risk in AMD-affected mine waste: mineralogical and geochemical perspectives. *Environ Earth Sci*. 2025;85:8. <https://doi.org/10.1007/s12665-025-12602-9>.
49. DeSantis MK, Triantafyllidou S, Schock MR, Lytle DA. Mineralogical Evidence of Galvanic Corrosion in Drinking Water Lead Pipe Joints. *Environ Sci Technol*. 2018;52:3365–74. <https://doi.org/10.1021/acs.est.7b06010>.
50. Seltenrich N. A Fix for Fixtures: Addressing Lead Contamination in West African Drinking Water. *Environ Health Perspect*. 2021;129:084003. <https://doi.org/10.1289/EHP9610>.
51. Lobo GP, Kalyan B, Gadgil AJ. Electrochemical deposition of amorphous aluminum oxides on lead pipes to prevent lead leaching into the drinking water. *J Hazard Mater*. 2022;423:127195. <https://doi.org/10.1016/j.jhazmat.2021.127195>.
52. Santucci RJ, Scully JR. The pervasive threat of lead (Pb) in drinking water: Unmasking and pursuing scientific factors that govern lead release. *Proc Natl Acad Sci*. 2020;117:23211–8. <https://doi.org/10.1073/pnas.1913749117>.
53. Cartier C, Arnold RB, Triantafyllidou S, Prévost M, Edwards M. Effect of Flow Rate and Lead/Copper Pipe Sequence on Lead Release from Service Lines. *Water Res*. 2012;46:4142–52. <https://doi.org/10.1016/j.watres.2012.05.010>.
54. Chu HTT, Vu TV, Nguyen TKB, Nguyen HTH. Accumulation of Arsenic and Heavy Metals in Native and Cultivated Plant Species in a Lead Recycling Area in Vietnam. *Minerals*. 2019;9. <https://doi.org/10.3390/min9020132>.
55. Lubiński K, Lener MR, Marciniak W, Pawłowski J, Sadzikowska J, Kiljańczyk A, et al. Blood Lead (Pb) Levels as a Possible Marker of Cancer Risk in a Prospective Cohort of Women with Non-Occupational Exposure. *Biomedicines*. 2025;13. <https://doi.org/10.3390/biomedicines13071587>.
56. Yu Y-L, An D-W, Yang W-Y, Zhang D-Y, Martens DS, Nawrot TS, et al. Health risks related to environmental and occupational lead exposure. *Pol Heart J Kardiologia Pol*. 2025;83:138–48. <https://doi.org/10.33963/v.phj.104575>.
57. Specht AJ, Lin Y, Xu J, Dickerson AS, Yan C, Hu H, et al. The effect of chelation on bone Pb stores in Pb poisoned children. *J Expo Sci Environ Epidemiol*. 2025;35:709–14. <https://doi.org/10.1038/s41370-024-00735-1>.
58. Mohammed RS, Mourad BH. The association between occupational lead exposure and serum levels of vitamin D3 and a bone turnover biomarker in smelter workers. *Int Arch Occup Environ Health*. 2025;98:297–307. <https://doi.org/10.1007/s00420-025-02125-y>.
59. Zabihi A, Mehrpour O, Nakhaee S, Atabati E. Lead poisoning and its effects on bone density. *Sci Rep*. 2025;15:15619. <https://doi.org/10.1038/s41598-025-92236-w>.
60. Yang S, Li J, Wu Y. The association of lead and cadmium exposure with periodontitis: a systematic review and meta-analysis. *BMC Oral Health*. 2025;25:935. <https://doi.org/10.1186/s12903-025-06195-9>.
61. Shao K, Yu Y, Ritz B, Paul KC. DNA methylation biomarkers for cumulative lead exposures and cognitive impairment. *Environ Res*. 2025;264:120304. <https://doi.org/10.1016/j.envres.2024.120304>.

62. Kootbodien T, Naicker N, Street R, Mathee A. Cumulative lead exposure and cognitive performance among the elderly in Johannesburg, South Africa. *Environ Adv*. 2025;21:100652. <https://doi.org/10.1016/j.envadv.2025.100652>.
63. Lalwani P, King DE, Morton KS, Rivera NA, Huayta J, Hsu-Kim H, et al. Increased cytotoxicity of Pb2+ with co-exposures to a mitochondrial uncoupler and mitochondrial calcium uniporter inhibitor. *Environ Sci Process Impacts*. 2023;25:1743–51. <https://doi.org/10.1039/D3EM00188A>.
64. Larsen B, Sánchez-Triana E. Global health burden and cost of lead exposure in children and adults: a health impact and economic modelling analysis. *Lancet Planet Health*. 2023;7:e831–40. [https://doi.org/10.1016/S2542-5196\(23\)00166-3](https://doi.org/10.1016/S2542-5196(23)00166-3).
65. Chitambwe T, Susanna D. Environmental Health Risks to Children in Sub-Saharan Africa: A Scoping Review of Evidence and Gaps in Global Interventions. *J Epidemiol Kesehatan Indones*. 2025;9. <https://doi.org/10.7454/epidkes.v9i3.1163>.
66. Clay K, Hollingsworth A, Severnini E. The Impact of Lead Exposure on Fertility, Infant Mortality, and Infant Birth Outcomes. *Rev Environ Econ Policy*. 2024;18:301–20. <https://doi.org/10.1086/730854>.
67. Wessel L. Prevention of Lead Exposure in the Perinatal Period. *J Obstet Gynecol Neonatal Nurs*. 2026;55:57–68. <https://doi.org/10.1016/j.jogn.2025.08.005>.
68. Garcia JZ, CCATE C de S, Ali N, Arango O, Brady C, Esquivel-Cote R, et al. A Community-Based Participatory Research Program to Assess Community Lead Exposure Risk: Establishing Research Priorities. *Res Sq*. 2025. <https://doi.org/10.21203/rs.3.rs-6279421/v1>. :rs.3.rs-6279421.

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